

Computer Vision SS26 - Exercise 1

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This exercise will be discussed the week of 4th of May.

1 Introduction

The following exercises cover digital images, homogeneous coordinates, sampling, and 2D geometric transformations.

2 Exercises

Exercise 1: Digital Images and Quantization

A grayscale image can be modeled as a function

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}.$$

A digital 8-bit grayscale image is instead represented on a finite discrete grid with discrete intensity values.

Let an image have resolution 2048×1536 and 8-bit grayscale intensities.

1. Define the domain and codomain of this digital image mathematically.
2. How many pixels does the image contain?
3. How many different intensity values can each pixel take?
4. Explain the difference between quantization of coordinates and quantization of intensities.

Exercise 2: Homogeneous Coordinates

Let a 2D point be given by

$$x = \begin{pmatrix} 4 \\ 2 \end{pmatrix}.$$

1. Write this point in homogeneous coordinates.

2. Show that

$$\tilde{x}_1 = \begin{pmatrix} 4 \\ 2 \\ 1 \end{pmatrix} \quad \text{and} \quad \tilde{x}_2 = \begin{pmatrix} 8 \\ 4 \\ 2 \end{pmatrix}$$

represent the same Euclidean point.

3. What is the geometric meaning of a homogeneous point with third coordinate $w = 0$?

Exercise 3: 2D Transformations

Consider the Euclidean point

$$x = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

It is first rotated by $\theta = 90^\circ$ around the origin and then translated by

$$t = \begin{pmatrix} 3 \\ 1 \end{pmatrix}.$$

1. Write the rotation matrix R_θ .
2. Compute the transformed point

$$x' = R_\theta x + t.$$

3. Write the corresponding rigid transformation matrix in homogeneous coordinates.
4. Explain why this transformation preserves distances.